Responsibility Convergence in Organizational Investigations:

Structural Dynamics of Blame in Root Cause Analysis

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Abstract

Organizational investigations such as root cause analysis (RCA) are intended to identify systemic causes of failure and support organizational learning. Yet across regulated and safety-critical environments, investigative conclusions recurrently stabilize responsibility at identifiable individuals despite formal commitments to system-oriented explanation. This persistent actor-centered pattern presents a theoretical puzzle: why do investigative systems designed to diffuse responsibility across sociotechnical structures so often concentrate accountability at particular organizational members?

This paper develops a structural account of this recurrent stabilization dynamic, termed responsibility convergence in organizational investigations. Responsibility convergence refers to the patterned concentration of accountability at identifiable actors when investigative processes translate causal ambiguity into administratively actionable conclusions. The framework specifies how convergence arises from the interaction of procedural ambiguity, attributional closure pressure, and accountability legibility bias within investigative reasoning. Under conditions of underdetermined causal reconstruction, organizations face institutional pressure to stabilize responsibility assignments. Because identifiable individuals constitute the most interpretable and administratively actionable attribution anchors within accountability infrastructures, attribution systematically concentrates on

actors rather than distributed structural conditions.

By situating investigative convergence within broader organizational responsibility alignment dynamics, the paper clarifies how RCA outcomes reflect not deviation from systemic ideals but the structural operation of accountability infrastructures. The analysis articulates the mechanisms of convergence, derives theoretical propositions, and outlines implications for organizational learning and accountability design.

1. Introduction

Organizational investigations are central mechanisms through which institutions interpret failures and stabilize responsibility attribution. In safety-critical and highly regulated domains, adverse events rarely have a single cause but emerge from interactions among technologies, procedures, and human action. Investigative frameworks such as root cause analysis (RCA) are designed to diffuse responsibility across sociotechnical systems and identify contributing factors across organizational levels rather than individual culpability (Reason, 1990). These approaches reflect a long-standing recognition in safety science and organizational theory that failures in complex systems are distributed phenomena requiring systemic explanation (Perrow, 1984; Vaughan, 1996).

Beyond their analytic function, investigations also play a constitutive organizational role. They are the processes through which ambiguous events are rendered intelligible, accountability is assigned, and lessons for future prevention are articulated. Across domains such as safety incidents, compliance failures, and operational disruptions, investigations operate as formalized episodes of organizational sensemaking in which causally underdetermined situations must be translated into administratively actionable accounts that stabilize responsibility (Weick, 1995). These processes are expected simultaneously to preserve systemic explanations and to produce determinate conclusions that support remediation, sanction, and learning. The tension between distributed causality and determinate accountability makes investigative settings revealing sites for examining responsibility

stabilization in practice.

However, across regulated and safety-critical environments, investigative structures tend to stabilize responsibility at identifiable actors such as operators, supervisors, or local decision makers, even when formal frameworks emphasize systemic causation. Accident inquiries, audit reviews, and compliance investigations recurrently channel corrective actions centered on retraining, procedural adherence, or disciplinary measures directed at specific individuals or roles (Dekker, 2007; Power, 1997). This convergence pattern persists despite explicit commitments to system-level explanation and organizational learning. The recurrent stabilization of accountability at identifiable actors therefore poses a theoretical puzzle: why do investigative systems designed to diffuse responsibility across sociotechnical structures so often stabilize attribution at particular organizational members?

This paper addresses this question by introducing the concept of responsibility convergence. Responsibility convergence refers to the structural stabilization of accountability at identifiable actors when investigative processes, structured by accountability infrastructures, translate causal ambiguity into administratively actionable conclusions. Rather than reflecting intentional blame or cognitive bias alone, convergence arises from structural properties of investigative reasoning and accountability infrastructures. The analysis develops a mechanism model in which procedural ambiguity, closure pressure, and legibility bias sequentially channel attribution toward actors who provide the most interpretable, documentable, and sanctionable anchors for organizational action.

Situating investigative convergence within broader responsibility dynamics, the paper contributes to organizational theory in three ways. First, it reframes blame in investigations as a predictable structural stabilization of responsibility rather than a deviation from systemic ideals. Second, it integrates attribution theory with accountability and governance perspectives by specifying how responsibility becomes operationally actionable within organizational procedures. Third, it positions responsibility convergence as the investigative stage within a broader research program on responsibility structuring across organizational processes, linking procedural pre-

configuration, anticipatory alignment, and post-hoc stabilization of accountability. Responsibility convergence operates at the post-event stage of organizational accountability processes, complementing anticipatory alignment dynamics described in anticipated accountability convergence (AAC), which structure expectations of accountable action prior to execution. Together, these perspectives situate responsibility structuring across the temporal arc of organizational action—from anticipatory alignment to investigative stabilization. Responsibility convergence complements prior work on cognitive friction by specifying how responsibility becomes stabilized after events within investigative infrastructures.

2. Responsibility Attribution in Organizational Investigations

Responsibility attribution in organizations has been examined across attribution theory, accountability research, and organizational blame studies (Heider, 1958; Weiner, 1985; Roberts, 1991). Attribution theory distinguishes between dispositional and situational explanations of events, while organizational research has shown that negative outcomes are often attributed to individuals under conditions of uncertainty or causal complexity (Staw, 1975; Ross, 1977). Accountability scholarship further demonstrates that expectations of evaluation shape decision and justification processes (Tetlock, 1985; Lerner & Tetlock, 1999). However, these perspectives primarily explain attribution as a cognitive or evaluative process rather than as an outcome structured by investigative systems.

In organizational investigations, procedures such as RCA integrate attributional reasoning with procedural governance (Rooney & Heuvel, 2004; Andersen & Fagerhaug, 2006). Investigators must construct causal narratives linking events, actions, and outcomes. Yet organizational failures rarely yield complete causal reconstruction, leaving causal relations structurally indeterminate (Perrow, 1984; Vaughan, 1996). Event sequences are distributed, nonlinear, and often partially documented. Multiple plausible explanations coexist, and evidentiary completeness is rarely

achievable.

This structural indeterminacy produces a closure requirement for accountability stabilization. Investigations cannot remain indefinitely ambiguous; they must stabilize conclusions to enable corrective action, regulatory documentation, and organizational learning processes (Dekker, 2007; Reason, 1997). Responsibility attribution in RCA contexts therefore occurs under conditions of constrained certainty and institutional pressure for determinate outcomes.

Within this context, responsibility convergence can be understood as a stabilization dynamic: attribution settles on targets that satisfy interpretive clarity and administrative actionability under ambiguity. Because organizational accountability expectations typically orient toward identifiable actors (Roberts, 1991; Power, 1997), investigative closure tends to settle on individuals even when causation is distributed. RCA procedures formalize this convergence dynamic by requiring determinate causal attribution and corrective assignment even when causal reconstruction remains distributed or indeterminate.

3. Mechanisms of Responsibility Convergence

The analysis identifies three interacting mechanisms that generate responsibility convergence in organizational investigations: procedural ambiguity, attributional closure pressure, and accountability legibility bias.

3.1 Procedural Ambiguity

Procedural ambiguity refers to indeterminacy in reconstructing causal sequences and accountability relations (Langley, 1999; Vaughan, 1996; Weick, 1995). Investigations rely on retrospective interpretation, incomplete documentation, and contested narratives. When causal chains cannot be definitively established, multiple attribution pathways remain viable. Responsibility assignments therefore remain indeterminate, generating pressure toward attributional convergence.

The interpretive indeterminacy introduced by procedural ambiguity does not remain neutral

within investigative processes. Because investigations are conducted under expectations of explanation and resolution, ambiguity generates pressure for investigative closure: the need to transform open-ended interpretive space into determinate accounts that can support organizational action. The more procedural pathways multiply or remain underspecified, the more strongly investigative reasoning is driven toward selecting among them. Ambiguity thus structurally amplifies the demand for closure by expanding the range of plausible interpretations that must ultimately be resolved into accountable conclusions.

3.2 Attributional Closure Pressure

Investigative processes require determinate conclusions to support corrective action, regulatory reporting, and organizational continuity (Power, 1997; March et al., 2000; Dekker, 2007). Attributional closure pressure arises from this requirement. As ambiguity persists, pressure increases to select an explanation that is defensible and actionable. Investigative closure transforms open causal interpretation into determinate attribution outcomes.

As investigative processes move toward resolution, closure pressure operates by structuring the selection environment of attribution. Among multiple plausible causal accounts, investigators are driven to privilege explanations that can sustain organizational scrutiny and support actionable decisions. This does not necessarily resolve causal ambiguity but constrains which interpretations remain viable within institutional requirements. Closure pressure thus narrows the interpretive field, channeling attribution toward determinate outcomes that can be justified, documented, and operationalized.

3.3 Accountability Legibility Bias

Accountability systems orient toward identifiable agents as attribution targets (Latour, 1987; Roberts, 2009). This orientation produces an accountability legibility bias: responsibility is more readily assigned to entities that are traceable, documentable, and sanctionable within organizational records and governance processes. While prior accountability research has emphasized identifiable agents,

this analysis specifies how legibility operates as a structural selection mechanism within investigative closure. Individual actions typically satisfy these criteria because they appear discrete and normatively evaluable, whereas systemic causes remain diffuse, distributed, and structurally embedded. Because accountability infrastructures privilege attribution targets that can be clearly recorded and acted upon, individuals become the most legible recipients of responsibility under conditions of attributional closure.

Closure pressure alone, however, does not determine where responsibility ultimately stabilizes. Investigative processes must produce conclusions that are not only determinate but also organizationally actionable. Closure therefore operates through attribution anchors that satisfy documentation, remediation, and accountability requirements, directing attribution toward actors readily specified within organizational records and corrective frameworks. Responsibility convergence thus reflects not merely a bias toward individual blame but the structural requirement that accountability systems render causality administratively actionable under evidentiary constraint.

3.4 Convergence Dynamics

These mechanisms interact. Procedural ambiguity structurally intensifies closure pressure; closure pressure in turn activates or amplifies legibility bias; legibility bias ultimately directs attribution toward identifiable actors (Weick, 1995; Dekker, 2007; Power, 1997). Responsibility convergence therefore emerges structurally from investigative reasoning under accountability constraints, without requiring intentional blame.

This dynamic also reflects broader organizational tendencies toward aligning responsibility with identifiable accountability targets. Investigative convergence thus constitutes a localized manifestation of broader responsibility alignment pressures within regulated accountability systems.

4. Propositions

The convergence model yields four theoretical propositions.

Proposition 1. Greater procedural ambiguity in causal reconstruction increases responsibility convergence toward identifiable actors.

As ambiguity in causal reconstruction expands, multiple attribution pathways remain viable and responsibility assignments remain indeterminate. Under such conditions, investigative processes face increasing demands to stabilize attribution. Because accountability systems orient toward identifiable agents as the most interpretable and actionable attribution targets, ambiguity intensifies the likelihood that responsibility ultimately converges on individuals rather than diffuse systemic causes.

Proposition 2. Attributional closure pressure strengthens the effect of procedural ambiguity on convergence toward identifiable actors.

Ambiguity alone does not determine where attribution converges; it produces a field of competing explanations. Closure pressure, however, requires the selection of a determinate and defensible account to enable organizational action. As closure pressure intensifies, investigative reasoning is increasingly driven to resolve ambiguity through attribution targets that can support corrective decision-making, thereby amplifying convergence toward identifiable actors.

Proposition 3. Accountability legibility bias mediates the relationship between attributional closure pressure and attribution to identifiable actors.

Closure pressure necessitates determinate attribution, but legibility determines which targets can satisfy this requirement. Accountability infrastructures privilege attribution targets that are traceable, documentable, and sanctionable within organizational records. Because individuals most readily meet these criteria, closure pressure is translated into actor-centered attribution through legibility bias, producing responsibility convergence at identifiable actors.

Proposition 4. Higher responsibility convergence toward identifiable actors is negatively associated with systemic learning depth in post-investigation corrective action processes (March et al., 2000).

When responsibility converges on identifiable actors, corrective processes tend to target

behavior, compliance, or local decision quality rather than underlying structural conditions. Because individuals provide the most visible and administratively tractable attribution anchors, organizational responses frequently emphasize retraining, monitoring, or discipline instead of redesigning procedures, resource configurations, or coordination structures. Actor-centered attribution directs investigative attention toward proximal actions, limiting examination of distributed system interactions underlying failure. Such actor-centered remediation narrows the scope of causal reconstruction and limits the exploration of systemic interactions that contributed to the event. As a result, investigations may achieve attributional closure without generating deeper organizational learning about structural vulnerabilities. Responsibility convergence thus shifts post-investigation adaptation toward local correction rather than systemic reconfiguration, reducing organizational learning depth even when formal learning processes exist.

Responsibility convergence is not inevitable. Convergence dynamics are likely to weaken in organizational contexts where accountability structures are explicitly distributed across collective units rather than oriented toward identifiable actors. In settings characterized by shared liability regimes, strong systemic transparency, and reduced sanction intensity at the individual level, attribution may remain more diffuse and structurally anchored. Under such conditions, closure pressure does not automatically translate into actor-centered attribution because accountability infrastructures themselves provide legible systemic anchors. Responsibility convergence therefore varies across institutional designs, depending on how accountability is structured and operationalized. When sanctionability is distributed across collective units rather than concentrated at identifiable actors, accountability infrastructures no longer privilege individual attribution as the most administratively efficient stabilization point.

5. Theoretical Contributions

The responsibility convergence framework contributes to organizational theory in three ways.

First, it reframes blame concentration as a structural convergence process rather than an intentional attributional choice. Individual attribution arises from investigative conditions rather than primarily reflecting attributional bias.

Second, the framework integrates attribution theory with accountability and governance literatures by specifying how procedural and institutional requirements shape attribution outcomes under conditions of causal ambiguity.

Third, it situates investigative attribution within broader organizational responsibility alignment dynamics. Organizational accountability systems tend to orient responsibility toward identifiable actors, and RCA conclusions reflect how localized investigative closure interacts with these wider alignment pressures. Investigative convergence thus complements anticipatory alignment dynamics by specifying how responsibility is stabilized after events within accountability systems.

This perspective clarifies why system-oriented investigative frameworks often produce person-centered outcomes. Convergence is not a deviation from investigative ideals but a predictable effect of ambiguity, closure, and legibility constraints within accountability infrastructures.

6. Implications for Organizational Learning and Accountability Design

Responsibility convergence has implications for both organizational learning and accountability infrastructures. When investigations concentrate responsibility on individuals, systemic vulnerabilities may be underdiagnosed and corrective actions may emphasize retraining or discipline rather than structural redesign. Over time, this pattern may sustain or reproduce organizational risk and erode perceptions of procedural fairness. This pattern aligns with the framework's prediction that responsibility convergence reduces systemic learning depth by directing corrective attention toward proximate actors rather than distributed structural conditions.

Mitigating convergence requires interventions mapped to its underlying structural mechanisms. Reducing procedural ambiguity through richer data capture and temporal

reconstruction tools may limit convergence pressure. Designing investigative protocols that tolerate provisional conclusions may reduce closure pressure. Enhancing the representation of systemic causal pathways may counteract legibility bias by making distributed causes more interpretable within accountability systems.

These interventions aim not to eliminate individual responsibility but to balance responsibility across levels, aligning investigative conclusions more closely with systemic causation.

7. Discussion: Investigative Closure and Responsibility Alignment

Organizational investigations translate ambiguous causal structures into stabilized responsibility assignments under accountability constraints. In regulated environments, where actions must be documented, justified, and auditable, attributional legibility becomes a central organizing principle of investigative reasoning. Responsibility convergence therefore reflects how investigative closure processes align with organizational tendencies to orient accountability toward identifiable actors.

Although developed in the context of root cause analysis, responsibility convergence is not unique to formal investigations. Similar convergence dynamics arise whenever organizations must translate causally underdetermined events into administratively actionable accounts, including performance evaluation, audit and compliance processes, and post-hoc decision justification. Across these contexts, distributed or structurally conditioned outcomes are routinely condensed into determinate conclusions that justify intervention. Organizational constraints, coordination breakdowns, or resource dependencies are frequently reframed as issues of individual competence, adherence, or judgment. In each case, responsibility convergence reflects the shared structural requirement that accountability be rendered identifiable and operational within organizational systems. The same underlying pressures recur: ambiguous causal reconstruction generates demands for closure, while accountability infrastructures privilege attribution targets that are traceable, recordable, and sanctionable. RCA investigations therefore provide an analytically visible site in

which a broader organizational tendency becomes concentrated and observable.

From this perspective, RCA processes do not merely discover causes; they also structure responsibility through interpretive stabilization. The concentration of attribution on individuals emerges from the interaction of ambiguity, closure pressure, and legibility bias within investigative contexts. These dynamics situate investigative blame within broader organizational patterns that orient responsibility toward identifiable accountability targets.

Responsibility convergence thus reveals how organizations transform distributed causality into administratively actionable attribution. Investigative conclusions represent stabilized points of responsibility alignment under uncertainty. This stabilization dynamic also shapes organizational sensemaking and learning, as attributional closure influences which causal pathways remain visible and actionable in subsequent corrective processes (Weick, 1995; March et al., 2000). Whereas cognitive friction identifies how procedural structures pre-configure localized judgment during task execution, responsibility convergence explains how investigative infrastructures subsequently stabilize responsibility attribution after events, locating responsibility alignment across the temporal arc of organizational action.

Responsibility convergence may therefore be understood as one phase within a broader temporal structuring of accountability, in which responsibility is shaped before action, stabilized during investigation, and subsequently embedded within organizational memory.

Future research could empirically examine responsibility convergence by analyzing patterns of attribution across formal investigative reports in regulated environments. Comparative analysis of root cause analysis documents, audit findings, or compliance investigations could assess whether greater procedural ambiguity and closure pressure are associated with increased concentration of responsibility at identifiable actors. Convergence may be operationalized through the distribution of attribution targets in investigative conclusions, corrective action assignments, and documentation practices. Such empirical work would allow systematic evaluation of when convergence intensifies,

when it weakens, and how institutional accountability architectures shape attribution outcomes across organizational contexts.

8. Conclusion

This paper introduces responsibility convergence as a structural explanation for the tendency of organizational investigations to attribute failures to individuals despite system-oriented intentions. By identifying procedural ambiguity, attributional closure pressure, and accountability legibility bias as interacting mechanisms, the framework clarifies how investigative processes translate causal indeterminacy into concentrated attribution.

Situating investigative convergence within broader organizational responsibility alignment dynamics highlights how RCA outcomes reflect both localized investigative closure and wider accountability structures. Recognizing convergence dynamics contributes to organizational accountability theory and informs the design of investigative systems that more accurately represent systemic causation.

Responsibility convergence thus provides a framework for understanding how organizations stabilize responsibility under uncertainty in regulated sociotechnical environments. Situated within a broader responsibility research program, the framework specifies how responsibility becomes stabilized during investigative attribution processes, complementing prior work on cognitive friction and anticipated accountability convergence. Whereas cognitive friction identifies how responsibility-bearing judgment is structurally pre-configured within procedural documentation, and anticipated accountability convergence explains how responsibility expectations align before participation, responsibility convergence clarifies how responsibility becomes stabilized after events through investigative closure. Together, these perspectives trace responsibility structuring across the temporal arc of organizational accountability systems—from anticipatory alignment to investigative stabilization and AI-mediated representation.

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